

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 10

Exercise 10.1 (Melnikov's Method).

a) Verify the equation

$$Df(\gamma)f(\gamma) \wedge \gamma_1^s + f(\gamma) \wedge (Df(\gamma)\gamma_1^s + g(\gamma, t)) = \text{tr}(Df(\gamma))\delta^s + f(\gamma) \wedge g(\gamma, t),$$

that was used in the proof of Theorem 14.5.

b) Assume that g does not depend on time and let $A \subset \mathbb{R}^2$ be the set enclosed by the homoclinic orbit γ .

Show that

$$M(t_0) = s \cdot \int_A \text{tr} Dg(x) dx \quad \text{with } s = 1 \text{ or } s = -1.$$

Solution.

a) We calculate

$$Df(\gamma)f(\gamma) = \begin{pmatrix} \partial_x f_1(\gamma) & \partial_y f_1(\gamma) \\ \partial_x f_2(\gamma) & \partial_y f_2(\gamma) \end{pmatrix} \begin{pmatrix} f_1(\gamma) \\ f_2(\gamma) \end{pmatrix} = \begin{pmatrix} \partial_x f_1(\gamma)f_1(\gamma) + \partial_y f_1(\gamma)f_2(\gamma) \\ \partial_x f_2(\gamma)f_1(\gamma) + \partial_y f_2(\gamma)f_2(\gamma) \end{pmatrix}$$

and likewise

$$Df(\gamma)\gamma_1^s = \begin{pmatrix} \partial_x f_1(\gamma)(\gamma_1^s)_1 + \partial_y f_1(\gamma)(\gamma_1^s)_2 \\ \partial_x f_2(\gamma)(\gamma_1^s)_1 + \partial_y f_2(\gamma)(\gamma_1^s)_2 \end{pmatrix}.$$

Hence we have

$$\begin{aligned} & Df(\gamma)f(\gamma) \wedge \gamma_1^s + f(\gamma) \wedge Df(\gamma)\gamma_1^s \\ &= (Df(\gamma)f(\gamma))_1(\gamma_1^s)_2 - (Df(\gamma)f(\gamma))_2(\gamma_1^s)_1 + f_1(\gamma)(Df(\gamma)\gamma_1^s)_2 - f_2(\gamma)(Df(\gamma)\gamma_1^s)_1 \\ &= (\partial_x f_1(\gamma)f_1(\gamma) + \partial_y f_1(\gamma)f_2(\gamma))(\gamma_1^s)_2 - (\partial_x f_2(\gamma)f_1(\gamma) + \partial_y f_2(\gamma)f_2(\gamma))(\gamma_1^s)_1 \\ &\quad + f_1(\gamma)(\partial_x f_2(\gamma)(\gamma_1^s)_1 + \partial_y f_2(\gamma)(\gamma_1^s)_2) - f_2(\gamma)(\partial_x f_1(\gamma)(\gamma_1^s)_1 + \partial_y f_1(\gamma)(\gamma_1^s)_2) \\ &= \partial_x f_1(\gamma)f_1(\gamma)(\gamma_1^s)_2 - \partial_y f_2(\gamma)f_2(\gamma)(\gamma_1^s)_1 + \partial_y f_2(\gamma)f_1(\gamma)(\gamma_1^s)_2 - \partial_x f_1(\gamma)f_2(\gamma)(\gamma_1^s)_1 \\ &= (\partial_x f_1(\gamma) + \partial_y f_2(\gamma))(f_1(\gamma)(\gamma_1^s)_2 - f_2(\gamma)(\gamma_1^s)_1) \\ &= \text{tr}(Df(\gamma))(f(\gamma) \wedge \gamma_1^s) \\ &= \text{tr}(Df(\gamma))\delta^s \end{aligned}$$

so it follows

$$Df(\gamma)f(\gamma) \wedge \gamma_1^s + f(\gamma) \wedge (Df(\gamma)\gamma_1^s + g(\gamma, t)) = \text{tr}(Df(\gamma))\delta^s + f(\gamma) \wedge g(\gamma, t).$$

Alternative: Using

$$v \wedge w = v^\top A w \quad \text{with } A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

we find

$$\begin{aligned} & Df(\gamma)f(\gamma) \wedge \gamma_1^s + f(\gamma) \wedge (Df(\gamma)\gamma_1^s + g(\gamma, t)) \\ &= f(\gamma)^\top Df(\gamma)^\top A \gamma_1^s + f(\gamma)^\top A Df(\gamma)\gamma_1^s + f(\gamma)^\top A \gamma_1^s \\ &= f(\gamma)^\top (Df(\gamma)^\top A + A Df(\gamma)) \gamma_1^s + f(\gamma) \wedge \gamma_1^s = (\star). \end{aligned}$$

A component-wise calculation shows that $B^\top A + AB = \text{tr}(B)A$ for all $B \in \mathbb{R}^{2 \times 2}$. This yields

$$\begin{aligned} (\star) &= \text{tr}(Df(\gamma))f(\gamma) \wedge \gamma_1^s + f(\gamma) \wedge \gamma_1^s \\ &= \text{tr}(Df(\gamma))\delta^s + f(\gamma) \wedge \gamma_1^s. \end{aligned}$$

b) We have

$$\begin{aligned} M(t_0) &= \int_{-\infty}^{\infty} f(\gamma(t - t_0)) \wedge g(\gamma(t - t_0)) dt \\ &= \int_{-\infty}^{\infty} f(\gamma(t)) \wedge g(\gamma(t)) dt \\ &= \int_{-\infty}^{\infty} f_1(\gamma(t))g_2(\gamma(t)) - f_2(\gamma(t))g_1(\gamma(t)) dt \end{aligned}$$

Since γ is a homoclinic orbit (and thus a solution) of the unperturbed equation we have $\gamma'(t) = f(\gamma(t))$ and hence we get

$$\begin{aligned} M(t_0) &= \int_{-\infty}^{\infty} \gamma'_1(t)g_2(\gamma(t)) - \gamma'_2(t)g_1(\gamma(t)) dt \\ &= \int_{-\infty}^{\infty} \left\langle \begin{pmatrix} g_1(\gamma(t)) \\ g_2(\gamma(t)) \end{pmatrix}, \begin{pmatrix} -\gamma'_2(t) \\ \gamma'_1(t) \end{pmatrix} \right\rangle dt \end{aligned}$$

Note that $\begin{pmatrix} -\gamma'_2(t) \\ \gamma'_1(t) \end{pmatrix}$ describes a normal vector to the orbit of γ . However, we do not know if it is pointing inwards or outwards (the correction term $s = \pm 1$ accounts for that). Furthermore, let

$$n = \frac{1}{\|\gamma'(t)\|} \begin{pmatrix} -\gamma'_2(t) \\ \gamma'_1(t) \end{pmatrix}$$

be the corresponding normal vector of unit length.

We rewrite the integral as a line integral and apply the divergence theorem (Gauss theorem) to obtain

$$\begin{aligned} M(t_0) &= \int_{-\infty}^{\infty} \langle g, n \rangle \|\gamma'(t)\| dt \\ &= \int_{\gamma} \langle g, n \rangle ds \\ &= s \cdot \int_A \text{div}(g) dx \\ &= s \cdot \int_A \partial_1 g_1(x) + \partial_2 g_2(x) dx \\ &= s \cdot \int_A \text{tr}(Dg(x)) dx. \end{aligned}$$

□

Exercise 10.2 (Sensitive dependence).

Let X be infinite and suppose that the map $f: X \rightarrow X$ is continuous and topologically transitive and the set of periodic points is dense in X .

Show that f has sensitive dependence on initial conditions on X with global sensitivity constant, i.e. there exists $r > 0$ s.t. for all $x \in X$, the definition of sensitive dependence on initial conditions applies.

Hint: Show first that for all open $U, V \subseteq X$, there exist $x \in U$, $n \geq 0$ s.t. $f^n(x) \in V$.

Solution.

We begin by proving that the dense forward orbit visits every open set infinitely often, which implies the hint: Let $V \subseteq X$ open, $(x_k)_{k \geq 0}$ dense orbit. By density of the periodic points, there exists a periodic point $x_p \in V$. Since the dense orbit cannot “hit” any periodic point, we have $x_p \notin \{x_k\}$. And since we also have $x_p \in \overline{\{x_k\}}$, we must have $|\{x_k\} \cap V| = \infty$.

We now solve the exercise. First observe that there is some $r > 0$ such that for all $x \in X$ there exists a periodic point $q \in X$ whose orbit is of distance at least $4r$ from x : Since X is infinite we can pick two periodic points with disjoint orbits and let $8r$ be the distance between the two orbits. By the triangle inequality the distance from at least one orbit to x must be at least $4r$.

Now fix some $x \in X$ and $\delta > 0$. Let q be a periodic point whose orbit is at distance at least $4r$. Without restriction we can assume $\delta < r$. Since the periodic points are dense, there is a periodic point $p \in B_\delta(x)$ of some period $n \in \mathbb{N}$.

Next, choose $\varepsilon > 0$ so small that $d(f^k(z), f^k(q)) < r$ for all $z \in B_\varepsilon(q)$ and all $k \in \{1, 2, \dots, n\}$.

Due to topological transitivity there is a $y \in B_\delta(x)$ which gets ε -close to q after N iterations. Now iterate another $k \leq n$ times such that $N + k$ is a multiple of n . By the choice of ε we have $d(f^{N+k}(y), f^k(q)) < r$. Consequently $f^{N+k}(y)$ is at least $3r$ away from x and since $f^{N+k}(p) = p$ lies in the δ -neighbourhood of x (and $\delta < r$), we know that $d(f^{N+k}(y), f^{N+k}(p)) > 2r$.

This means that either

$$d(f^{N+k}(x), f^{N+k}(y)) > r \quad \text{or} \quad d(f^{N+k}(x), f^{N+k}(p)) > r,$$

which shows sensitive dependence.

□

Exercise 10.3 (Chaos and conjugacies).

Let (X, ϕ_t) and (Y, ψ_t) be two dynamical systems that are topologically conjugate.

- Show that topological transitivity is preserved under topological conjugacy, i.e. (X, ϕ_t) is topologically transitive if and only if (Y, ψ_t) is topologically transitive.
- Now let X and Y be closed intervals (or more general: compact sets). Show that (X, ϕ_t) has sensitive dependence on initial conditions on X if and only if (Y, ψ_t) has sensitive dependence on initial conditions on Y .

Hint: Use uniform continuity.

Solution.

- a) Let (X, ϕ_t) and (Y, ψ_t) be two topologically conjugate dynamical systems, i.e. there is some $h : X \rightarrow Y$ such that $h \circ \phi_t = \psi_t \circ h$. Assume that (X, ϕ_t) is topologically transitive and let $\{\phi_t(x)\}_{t \in \mathcal{T}^+}$ be a dense (forward) orbit. Due to the conjugacy $\{h(\phi_t(x))\}_{t \in \mathcal{T}^+} = \{\psi_t(h(x))\}_{t \in \mathcal{T}^+}$ is an orbit of (Y, ψ_t) .

It remains to show that this orbit is dense in Y . Let V be some arbitrary open subset of Y . Since h is continuous, the preimage $h^{-1}(V) =: U$ is an open subset of X . As $\{\phi_t(x)\}_{t \in \mathcal{T}^+}$ is dense in X , there is some $t_0 \in \mathcal{T}^+$ such that $\phi_{t_0}(x) \in U$. This implies that $\psi_{t_0}(h(x)) = h(\phi_{t_0}(x)) \in V$ and consequently $\{\psi_t(h(x))\}_{t \in \mathcal{T}^+}$ is dense in Y .

- b) If X, Y are compact, the conjugating homeomorphism $h : X \rightarrow Y$ is uniformly continuous. That means for every $s > 0$ there is some $r > 0$, such that

$$|x_1 - x_2| < r \quad \Rightarrow \quad |h(x_1) - h(x_2)| < s$$

or equivalently:

$$|h(x_1) - h(x_2)| \geq s \quad \Rightarrow \quad |x_1 - x_2| \geq r. \quad (10.1)$$

Now suppose that (Y, ψ_t) has sensitive dependence on initial conditions (s.d.i.c.) with separation constant s . We show that (X, ϕ_t) has s.d.i.c. with separation constant r .

Take some $x_0 \in X$, $\delta > 0$ and denote $h(x_0) = y_0$. Since h^{-1} is continuous there is some $\varepsilon > 0$ such that $|h^{-1}(y_0) - h^{-1}(y_1)| < \delta$ for all y_1 with $|y_0 - y_1| < \varepsilon$.

Because (Y, ψ_t) has s.d.i.c. we know that there are $T > 0$ and $y_1 \in Y$ with $|y_0 - y_1| < \varepsilon$ such that

$$|\psi_T(y_0) - \psi_T(y_1)| \geq s.$$

As ϕ_t and ψ_t are conjugate, that means

$$|h(\phi_T(h^{-1}(y_0))) - h(\phi_T(h^{-1}(y_1)))| = |\psi_T(y_0) - \psi_T(y_1)| \geq s.$$

Now using the estimate (10.1) from above, we obtain

$$|\phi_T(h^{-1}(y_0)) - \phi_T(h^{-1}(y_1))| = |\phi_T(x_0) - \phi_T(h^{-1}(y_1))| \geq r.$$

To conclude we found $x_1 := h^{-1}(y_1)$ and $T > 0$ with $|x_1 - x_0| < \delta$ such that $|\phi_T(x_0) - \phi_T(x_1)| \geq r$, i.e., we have shown s.d.i.c. for (X, ϕ_t) .

□

Exercise 10.4 (Circle rotation).

On the circle $\mathbb{S}^1$, represented by the unit interval, consider the map

$$f : [0, 1) \rightarrow [0, 1), \quad x \mapsto x + \alpha \pmod{1}$$

for some fixed $\alpha \in \mathbb{R}$.

- a) Show that the dynamical system induced by iteration of f has periodic orbits if and only if $\alpha \in \mathbb{Q}$.
- b) Show that the system is topologically transitive if and only if $\alpha \in \mathbb{R} \setminus \mathbb{Q}$.

Hint: Show first that for every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that $d(x, f^n(x)) < \varepsilon$ for all $x \in [0, 1)$.

c) Is the system chaotic?

Solution.

- a) Note that f^n is given by $f^n(x) = x + n\alpha \bmod 1$. There is a n -periodic orbit (for some $n \in \mathbb{N}$) if and only if there is $x \in [0, 1)$ such that $f^n(x) = x + n\alpha \bmod 1 = x$. This means, there is $m \in \mathbb{N}$ such that $x + n\alpha = x + m$, which is equivalent to $\alpha = \frac{m}{n}$, i.e. $\alpha \in \mathbb{Q}$.
- b) Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and $\varepsilon > 0$ and take $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. The $N + 1$ points $x, f(x), \dots, f^N(x)$ are all distinct, since otherwise we would have a periodic orbit, in contradiction to what we proved in a). Since the unit interval has only length 1, there must be two of these points that are less than $\frac{1}{N}$ apart of each other, i.e. there are $m, n \in \mathbb{N}, n > m$, such that $d(f^m(x), f^n(x)) < \varepsilon$. Since f maintains the distance* of two points, we can conclude that also $d(x, f^{n-m}(x)) < \varepsilon$. Even more, $f^{n-m}(x) = x + \beta$ for some β with $|\beta| < \varepsilon$.

With this observation we can now easily show that even every orbit is dense. Let $x \in [0, 1)$ be arbitray and let U be some open interval in $[0, 1)$ of diameter ε . Since $d(x, f^{n-m}(x)) < \varepsilon$, it follows that there is some $k \in \mathbb{N}$, such that $f^{k(n-m)}(x) \in U$, which shows the density of the orbit of x , i.e. the dynamical system is topologically transitive.

If $\alpha \in \mathbb{Q}$, all orbits are periodic, so we cannot get an orbit that is dense in $[0, 1)$, i.e. the system is not topologically transitive.

- c) Since $d(f(x), f(y)) = d(x, y)$, we have no sensitive dependence on initial conditions and hence no chaos.

Remark: Choosing the “normal” distance on the interval (ignoring the periodicity/topology of the circle) we can get sensitive dependence. But that also makes the map f discontinuous.

□

* The distance on the unit interval (representing $\mathbb{S}^1$) is here not given by $d(a, b) = |b - a|$, but $d(a, b) = \min\{|b - a|, 1 - |b - a|\}$ using the topology of the circle.